

Introduction to Variant Annotation



TOPICS COVERED

- What is Variant Annotation?
- Exonic and Intronic Variants
- Databases and Annotation Tools
- Allele Frequencies
- Key points for variant assessment
- Clinical Significance and Reporting

What is Variant Annotation?

Assigning functional information to DNA variants

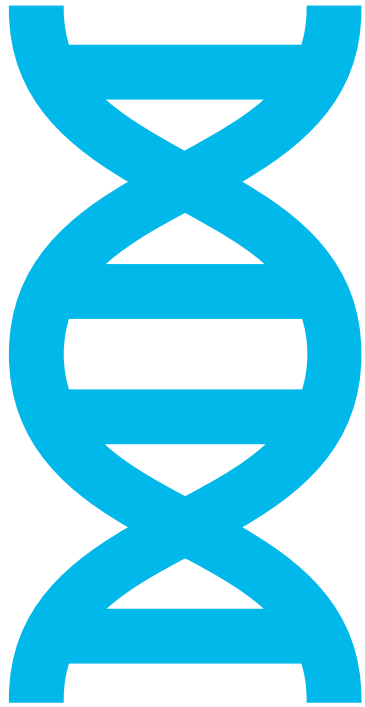
Describe the nature and the effect of the genomic aberrations produced by a variant

Importance:

- prioritize and filter genetic variants
- distinguish between benign and potentially pathogenic variations
- understanding of population genetics, evolution, and the regulatory networks governing gene expression.

Exonic- Intronic Variants

- Exonic variants are often scrutinized for their potential role in contributing to diseases. Intronic variants may affect gene expression or splicing, contributing to disease susceptibility.
- contribute to the genetic diversity within populations.



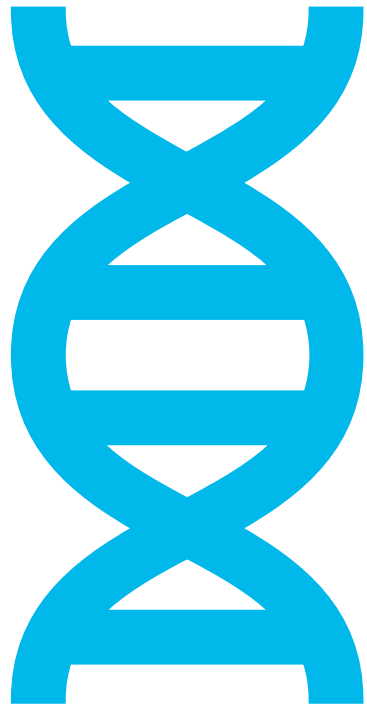
DATABASES AND ANNOTATION TOOLS

Population Databases

Information on the frequency of genetic variants in large, diverse populations.

Examples:

- [gnomAD](#)
- [Exome Aggregation Consortium](#)
- [1000 Genomes Project](#)
- [dbSNP](#)
- [dbVar](#)

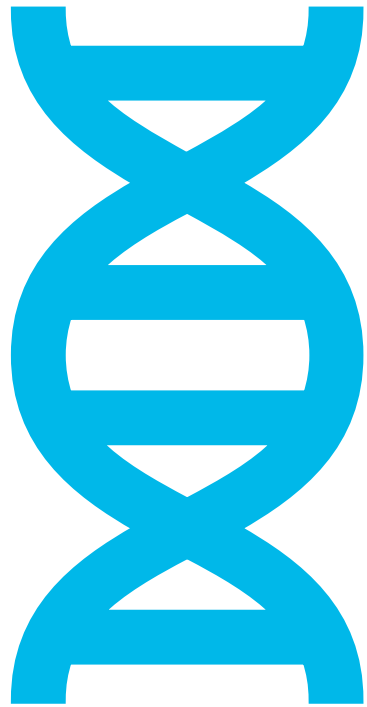


Disease-Specific Databases

Contain variants found in patients with diseases and assessments of variant pathogenicity.

Examples:

- [ClinVar](#)
- [OMIM](#) (Online Mendelian Inheritance in Man)
- [Human Gene Mutation Database](#)
- [DECIPHER](#)
- [HGVS](#) (Human Genome Variation Society)
- [LOVD](#) (Leiden Open Variation Database)



Sequence Databases

Include full genome sequences, gene reference sequences, and other genomic information.

Examples:

- [NCBI Genome](#)
- [RefSeqGene](#)
- [Locus Reference Genomic \(LRG\)](#)
- [MitoMap](#)
- [HumanMitoSeq](#)

TOOLS FOR VARIANT ANNOTATION

These tools use various methods, including evolutionary conservation, protein structure analysis, and splice site prediction, to provide insights into the potential impact of genetic variants on biological function

SIFT

MAPP

ANNOVAR

Ensembl
VEP

snpEff

CADD

Polyphen

FSPLICE

PhyloP

gnomAD Allele Frequencies (AFs)

Understanding allele frequencies is crucial for distinguishing between common and rare variants, guiding variant's potential pathogenicity in different population

Overview of gnomAD Database:

- gnomAD aggregates genomic data from diverse populations, providing a comprehensive catalog of allele frequencies. It is a valuable resource for researchers studying normal genetic variation.

Interpreting Allele Frequency Information:

- Researchers use gnomAD to assess the prevalence of variants in populations. Rare variants may be more likely to cause diseases, while common variants may have functional significance.

Practical Examples:

- In clinical and research settings, gnomAD data helps researchers contextualize variant frequencies, aiding in the interpretation of novel and rare variants.

gnomAD AF, VAF, MAF

Allele Frequency in GnomAD (AF_{gnomAD}):

- Formula: $AF_{gnomAD} = \text{Allele count} / \text{Number of alleles}$
- represents the frequency of a specific variant in a population within the gnomAD database.
- The allele count is the number of occurrences of the variant, and the number of alleles is the total number of times the variant was called.

Variant Allele Frequency/Fraction (VAF):

- VAF is the percentage of sequence reads matching a specific DNA variant divided by the overall coverage at that locus in a particular sample.
- It is a property of a variant within an individual sample.

Minor Allele Frequency (MAF) and Major Allele Frequency (mAF):

- mAF is the frequency of the less frequent allele in a population.
- MAF can be used interchangeably with mAF, which stands for minor allele frequency.
- Major Allele Frequency (MAF) is the frequency of the more common allele in a population.



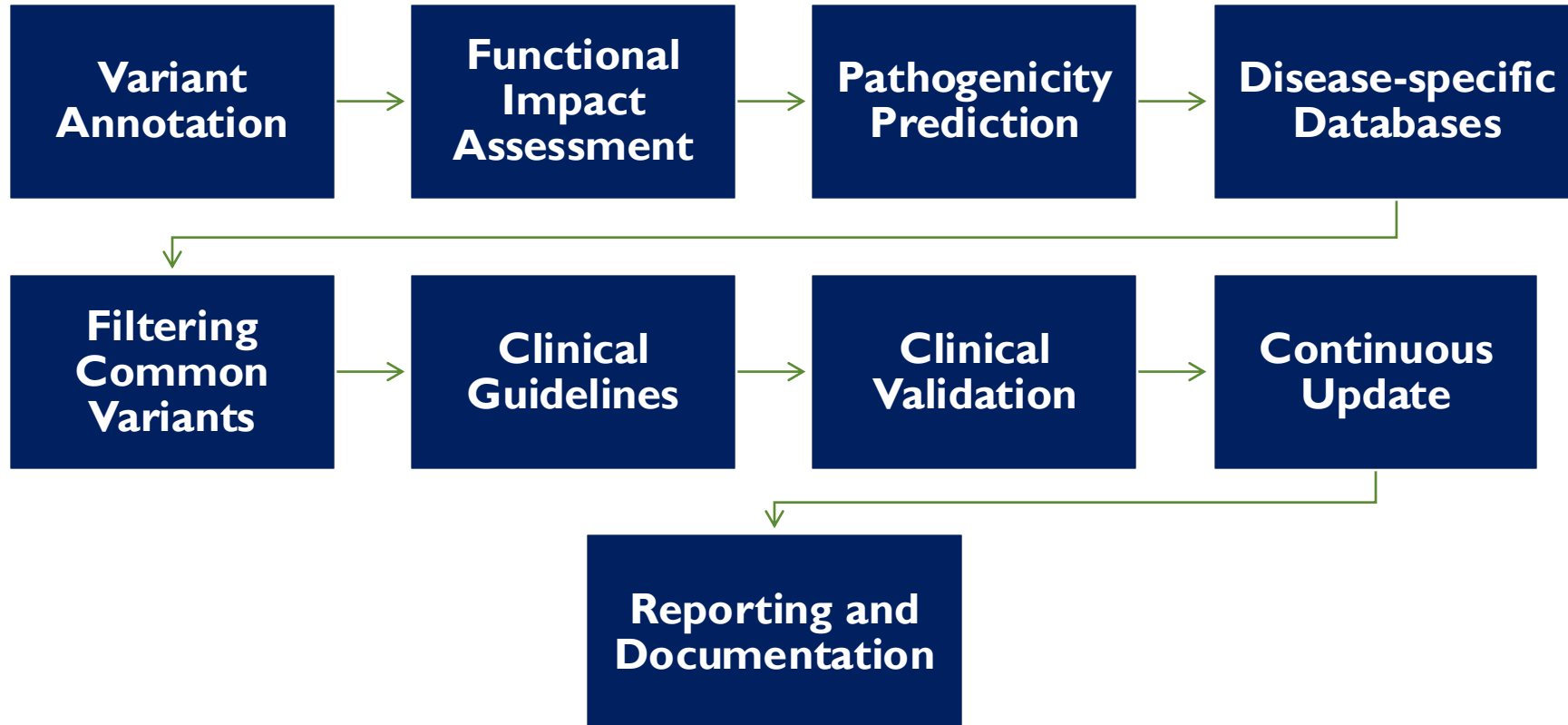
KEY POINTS FOR VARIANT ASSESSMENT

- **Functional Impact**
- **Population Frequency**
- **Reporting Guidelines**
- **Caution with Older Literature**
- **Privacy considerations while using Commercial Databases**
- **Collaboration with Clinicians**



Clinical Significance and Reporting

How clinically relevant variants are identified from annotated variants?





QUESTIONS